

About DONG-IN ENSIS

After 35 years of building control panels, we are turning them into data infrastructure.

Since 1991, DONG-IN ENSIS has designed and built control panels for industrial and marine sites. On that foundation, we now make Intelligent Control Panels that collect and connect equipment data while controlling it.



FIG. 01 Connecting field equipment data · illustrative video

35⁺

YEARS

Founded 1991

750⁺

PROJECTS

Projects delivered

15⁺

INDUSTRIES

Industries served

40⁺

CE·UL CERTIFIED

Certified solutions



OT – IT connection · The Intelligent Control Panel links field equipment data to IT (staged video)

01 · CEO Message

Industrial DX and AX start with field data.



Thank you for visiting DONG-IN ENSIS.

We began in 1991 as a supplier of industrial control equipment and grew into a company that designs and builds control systems for ships, energy facilities and plants. Over more than 750 projects, one thing became clear: **most equipment is already controlled by PLCs and drives, but the data they generate never leaves the control panel.**

AI and smart factories need field data first. Our Intelligent Control Panel collects data where the equipment is controlled and delivers it to the cloud in a standard format. For manufacturers, it is the first step toward managing equipment with data. For IT companies, it is a reliable source of field data.

We will keep adding data and intelligence to the control technology we have built in the field for 35 years.

CEO, DONG-IN ENSIS CO., LTD. **Min-gi Baek**

1991

Industrial control equipment supply

Global power and control brands



750⁺

Projects delivered

Marine, energy and plant control systems



2026

Intelligent Control Panel

From the control panel to the cloud

Company overview

Company

DONG-IN ENSIS CO., LTD.

(주)동인엔시스

CEO

Min-gi Baek

Founded

October 1991

Incorporated July 1997

Business

Intelligent Control Panels · industrial and marine control system engineering and manufacturing · system integration · supply of global power and control products

Sites

Head office 47, Jinyeon-ro 9beon-gil, Busanjin-gu, Busan

R&D center 8F NS Tower, 47, Jinyeon-ro 9beon-gil, Busanjin-gu, Busan · est. 2020

Yangsan Factory 1 107, Gongwon-ro, Sangbuk-myeon, Yangsan-si, Gyeongsangnam-do

Yangsan Factory 2 103-5, Gongwon-ro, Sangbuk-myeon, Yangsan-si, Gyeongsangnam-do

Gijang Factory 20, Dareumsan-gil, Jeonggwan-eup, Gijang-gun, Busan

Management systems

ISO 9001

ISO 14001

ISO 45001

Partners

Schneider Electric official SI partner

ABB

Fuji Electric

Eaton

Danfoss



Port container crane · staged image

VISION

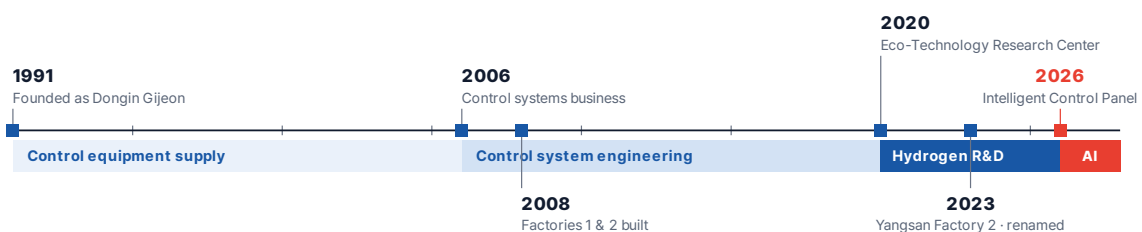
Accelerating DX and AX in manufacturing and maritime industries with **field equipment data**.

MISSION

Creating and connecting **reliable data** from the control panels that run the equipment.

From components to systems, from systems to data

35 years of growth, drawn to the actual timeline scale.



1991 ~

Control equipment supply

Founded as Dongin Gijeon.
Supplying global brands such as Siemens, Fuji Electric, ABB and Danfoss

2006 ~

Control system engineering

Industrial and marine control systems: design, build, software and commissioning

2020 ~

Green ship & hydrogen R&D

Eco-Technology Research Center founded; national R&D projects and liquid hydrogen station control

2026 ~

Intelligent Control Panel

Panel-based industrial data infrastructure connecting field equipment to cloud and AI

04 · Why We Can

Three foundations behind the Intelligent Control Panel

Data infrastructure is not built with software alone. It starts with hands-on experience controlling real equipment.



01 · MANUFACTURING

We design and build in-house

From panel design to build, wiring, PLC programming, factory testing and on-site commissioning.

Yongsan Factories 1 & 2 (completed 2023) and Gijang Factory

Dedicated FAT test line

Design and build to CE-UL standards



02 · FIELD EXPERIENCE

We control demanding sites

We have built and verified control systems in hydrogen, shipbuilding and plant sites with strict safety requirements.

Liquid hydrogen station control systems nationwide

FGSS test facility control for a leading shipbuilder

Hybrid electric propulsion control for green ships



03 · R&D

We research data and predictive control

Our Eco-Technology Research Center leads national and regional R&D and industry-academia collaboration.

5 registered patents on liquid hydrogen station control systems

National R&D on safety-based hydrogen-powered vessels

Joint research and technology transfer with Korea Maritime & Ocean University

We design and build in-house

Panels are built in our own factories, and PLC/drive programming and commissioning are done by our engineers.

References in demanding sites

Liquid hydrogen charging stations for a major Korean energy company, FGSS test facilities for a leading shipbuilder, and hybrid electric propulsion control for eco-friendly vessels.

Global brand partner

Official Schneider Electric SI partner and supply partner for ABB, Fuji Electric, Eaton and Danfoss.

* Photos are illustrations.

From control to cloud, one control panel.

DONG-IN ENSIS Intelligent Control Panels control your equipment with a PLC while collecting its operating data, and connect it securely to the cloud without extra data-collection hardware.

35+ years

of OT engineering

PLC → Cloud

no extra gateway

MQTT / TLS

data transfer · secured link

REST API

MES · monitoring · analytics
integration

Background: liquid hydrogen station · illustration

05 · Intelligent Control Panel

The control panel becomes data infrastructure

We reduced the usual PLC – gateway – server stack to a single control panel.

01

Real-time PLC control and data collection

Control logic and interlocks stay with the PLC on site.

02

MQTT / TLS cloud transfer

The PLC sends operating data over MQTT, with the link protected by TLS.

03

DataHub standard data · time-series storage

Data from each piece of equipment is organized in a common structure and stored over time.

04

REST API integration

A REST API your MES, monitoring and analytics systems can connect to.

From field equipment to AI, one continuous chain

HOW IT WORKS

Field

Equipment & sensors

Motors · pumps · drives
Temperature · vibration · power



Panel

Intelligent Control Panel

Real-time PLC control
Data collection
No extra gateway



TLS

MQTT / TLS

Encrypted transfer
Direct cloud link



Hub

DataHub

Standard data model
Time-series storage



API

Applications

Dashboards · alarms
AI · digital twin · MES

Control stays on site

The server never controls equipment directly; interlocks and final control remain with the PLC.

Secure connection

Field and cloud are linked through TLS-encrypted MQTT.

A data model that scales

As equipment and sensors are added, AI and digital twins keep using the same standard structure.

No extra data-collection device needed to get your data

Control stays on site; operating data goes from the control panel to the cloud.

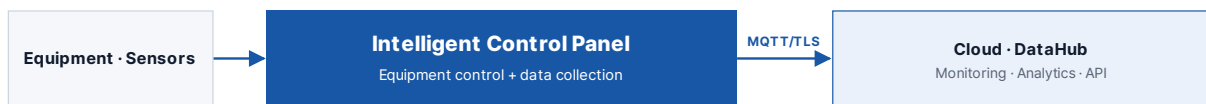
Typical setup vs. Intelligent Control Panel

WHY NO EXTRA DEVICE

Typical data-collection setup



DONG-IN ENSIS Intelligent Control Panel



▲ No extra device

Control stays on site; operating data goes from the control panel to the cloud. * Network and security configuration may vary depending on the site environment.

From land to sea, wherever motors run

One Intelligent Control Panel per site. We bring the control experience built in hydrogen, shipbuilding and plant projects.



* Application scenarios. Images are illustrations.

Most field problems start with missing data

Most equipment is already controlled by PLCs and drives, yet the data they produce never leaves the control panel.

■ ISSUE 01

Equipment stops at dawn, and no one knows why

Crews respond only after a failure, and lost production grows with every hour of downtime.

■ ISSUE 02

When experts retire, their know-how leaves too

Judgment based on sound and experience is never recorded.

■ ISSUE 03

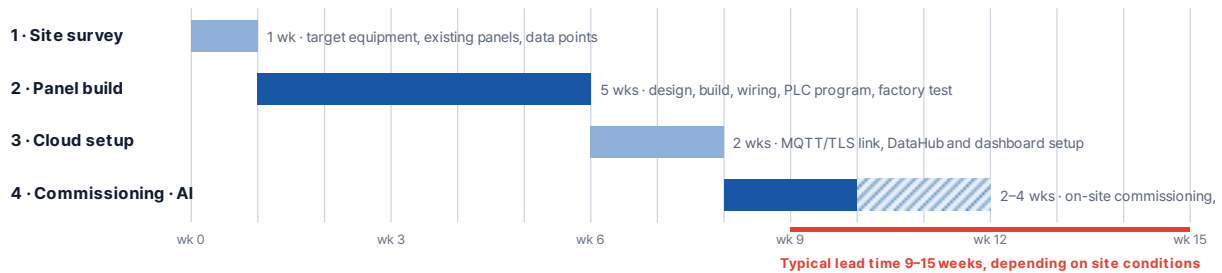
Signs of overheating and overload go unrecorded

Without a history of temperature, current and vibration, prevention is difficult.

The Intelligent Control Panel records data while it controls, and flags anomalies early.

From site survey to AI training, one company end to end

PROCESS · TYPICAL LEAD TIME 9–15 WEEKS



2026 · NOW

Data infrastructure

PLC → cloud → DataHub → dashboard, AI anomaly detection PoC

2027

Predictive maintenance

Anomaly detection and predictive maintenance on real data, digital twin link

NEXT

Smart factory expansion

Integrated control of more field equipment, including collaborative robots

We take care of the OT scope in your smart factory projects

If your AI and platform are ready but field data is not, DONG-IN ENSIS handles everything from the control panel to a standard API.

IT PARTNER · PLATFORM
· SERVICES

MES · ERP · EAM integration

AI analytics · predictive models

Digital twins · dashboards

Customer-facing platform operation

Standard data model + REST API · WebSocket

DONG-IN ENSIS · FIELD
DATA · OT

Panel design · build · installation · commissioning

PLC · drive · sensor integration

MQTT/TLS cloud transfer · DataHub

We focus on **connecting PLCs to the cloud as simply and securely as possible**. For SCADA, MES, AI and smart factory platforms, keep using your own solutions.

Key milestones since 2020

Since founding the Eco-Technology Research Center, we have expanded hydrogen and marine R&D and production capacity, leading to the Intelligent Control Panel business.

2026	Launched the Intelligent Control Panel business Selected for Innovation Premier 1000 (Ministry of Trade, Industry and Energy) · Registered as a Hydrogen Specialized Company · 5 patents registered on liquid hydrogen station control systems
2025	ISO 45001 certified MAINBIZ · INNOBIZ certified · selected as Preliminary Hydrogen Specialized Company
2024	Busan Strategic Industry Leading Company Regional Specialized Legend 50+ · family company agreement with Korea Maritime & Ocean University
2023	Yongsan Factory 2 completed · renamed DONG-IN ENSIS Joined the Ministry of Oceans and Fisheries project on small hydrogen-powered vessel technology and demonstration · Busan Hidden Champion; Service Strong SME
2022	National Representative Innovative Company (Ministry of Oceans and Fisheries) Busan Hydrogen Alliance member · Technology Credit Bureau (TCB) T-4 rating
2021	Electric hybrid fishing vessel project Joined the Ministry of Oceans and Fisheries project on core equipment for electric hybrid fishing vessels
2020	Eco-Technology Research Center founded Danfoss drives & soft starters and ABB low-voltage distribution agency

Milestones from 1991 to 2019 are shown in 03 · Our Journey.

Certifications · designations

ISO 9001 Quality management	ISO 14001 Environmental management	ISO 45001 Occupational health & safety	Corporate R&D Center Eco-Technology Research Center
INNOBIZ · MAINBIZ · Venture Ministry of SMEs and Startups	National Representative Innovative Company Ministry of Oceans and Fisheries	Innovation Premier 1000 MOTIE · 2026	Regional Specialized Legend 50+ Participating company
Strategic Industry Leading Company Busan Metropolitan City	Hidden Champion · Service Strong SME Busan Metropolitan City	Hydrogen Specialized Company Registered 2026	5 registered patents Liquid hydrogen station control systems

Location

HEAD OFFICE	R&D CENTER	FACTORY 1	FACTORY 2	FACTORY 3
Head office 47, Jinyeon-ro 9beon-gil, Busanjin-gu, Busan, Korea	Eco-Technology Research Center 8F NS Tower, 47, Jinyeon- ro 9beon-gil, Busanjin-gu, Busan	Yangsang Factory 1 107, Gongwon-ro, Sangbuk-myeon, Yangsan- si, Gyeongsangnam-do	Yangsang Factory 2 103-5, Gongwon-ro, Sangbuk-myeon, Yangsan- si, Gyeongsangnam-do	Gijang Factory 20, Dareumsan-gil, Jeonggwan-eup, Gijang- gun, Busan

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**Talk to us about adopting the Intelligent Control Panel
or partnering with us.**



**Learn more
on our website
donginensis.com**

PLANTS AND VESSELS

Will it work on your equipment?

Start with a site assessment.

SMART FACTORY · AI · MES COMPANIES

Need field data for your project?

Work with us as your OT partner.

CONTACT

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WEB

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From control to data, from data to
intelligence.

This company profile is based on the content of the DONG-IN ENSIS website. Photos and videos are illustrations.